Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 33

The Pigeonhole Principle

Covers Chartrand & Zhang, Section 8.3

Status of this set

Ungraded practice. Solutions released with it. For every problem, **name the pigeons and the holes explicitly** — that is the graded part.

Part I: How to Practice This Material

Three things to identify

1. The **pigeons** — the objects being placed.
2. The **holes** — the categories.
3. The **rule** assigning each pigeon to a hole.

Then: n pigeons in k holes forces some hole to hold at least $\lceil n/k \rceil$. To guarantee r in some hole you need $k(r-1)+1$ pigeons.

Choosing the holes is the creative step

The holes are rarely handed to you. Group the possible values so that any two items landing in the same group have the property you want — consecutive, summing to a target, sharing a remainder.

Three mistakes to expect

1. **Not naming the holes.** "By pigeonhole, two must match" is not an argument.
2. **Off-by-one on $k(r-1)+1$.** With $k(r-1)$ items the conclusion can fail.
3. **Claiming to know *which* two.** Pigeonhole proves existence, not identity.

Part II: Practice Problems

Section A — Basic counts

1. Answer each, naming the pigeons and holes.

(a) A drawer holds socks of 3 colours. How many must you draw to guarantee a matching pair?

(b) In a class of 30, how many must share a birth month?

(c) How many cards must you draw from a standard deck to guarantee 4 of the same suit?

2. Show that among any 8 integers, two of them differ by a multiple of 7.

Section B — Choosing the holes

3. Show that if 6 integers are chosen from $\{1, 2, \ldots, 10\}$, then two of them sum to 11.

4. Show that if 7 integers are chosen from $\{1, 2, \ldots, 12\}$, then two of them sum to 13. Then show 6 is not enough.

Section C — Writing

5. In five to eight complete sentences, explain why the pigeonhole principle is a *non-constructive* method. Use one of the problems above, and connect your answer to Lecture 26.

Section D — Challenge (optional)

6. Show that in any group of $n \geq 2$ people, two people have the same number of acquaintances within the group. (Assume acquaintance is mutual.)